2 Team Round Answers

Answer 1. 14400

Answer 2. 34

Answer 3. -7

Answer 4. $25\sqrt{3}$

Answer 5. 12

Answer 6. 7711

Answer 7. $\sqrt{130}$

Answer 8. 16

Answer 9. 17

Answer 10. $\sqrt{2}-1$

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3 Team Round Solutions

Problem 1. A six-digit natural number is "*sort-of-decreasing*" if its first three digits are in strictly decreasing order and its last three digits are in strictly decreasing order. For example, 821950 and 631631 are sort-of-decreasing but 853791 and 911411 are not. Compute the number of sort-of-decreasing six-digit natural numbers.

Solution 1. If three distinct digits are chosen from the set of digits $\{0, 1, 2, \ldots, 9\}$, then there is exactly one way to arrange them in decreasing order. There are $\binom{10}{3} = 120$ ways to choose the first three digits and 120 ways to choose the last three digits. Thus the answer is $120 \cdot 120 = \mathbf{14400}$.

Problem 2. For each positive integer N, let $P(N)$ denote the product of the digits of N. For example, $P(8) = 8$, $P(451) = 20$, and $P(2023) = 0$. Compute the least positive integer n such that $P(n + 23) = P(n) + 23$.

Solution 2. One can verify that no single-digit positive integer n satisfies the conditions of the problem.

If n has two digits, then $n + 23$ cannot be a three-digit number; this can be verified by checking the numbers $n \geq 88$, because if $n < 88$, then one of the digits of $n + 23$ is 0. Therefore both n and $n + 23$ must be two-digit numbers, so the only possible carry for $n + 23$ will occur in the tens place. If there is a carry for $n + 23$, then $n = \underline{a}\,\underline{8}$ or $n = \underline{a}\,\underline{9}$, while $n + 23 = \underline{(a+3)}\,\underline{1}$ or $n + 23 = \underline{(a+3)}\,\underline{2}$, respectively (the case $n = \underline{a}\,\underline{7}$ is omitted because then $P(n + 23) = 0$). In either case, $P(n + 23) < P(n)$ because $a \geq 1$. Otherwise, assume $n = \underline{a}\,\underline{b}$ and $n + 23 = \underline{(a+2)}\,\underline{(b+3)}$ is a solution to the given equation, which implies

$$23 = P(n + 23) - P(n) = (a + 2)(b + 3) - ab = 3a + 2b + 6.$$

This means $3a + 2b = 17$, which has solutions $(a, b) = (5, 1)$, $(3, 4)$ as a, b are digits and $b < 7$. The two-digit solutions are $n = 34$ or $n = 51$; thus the least n such that $P(n + 23) = P(n) + 23$ is $n = \mathbf{34}$.

Note: The following shows how to determine all solutions to the given equation. The reader may find this an interesting extension of the conditions of the problem.

Suppose that $n = \underline{a_1}\,\underline{a_2}\,\ldots\,\underline{a_k}$, which implies $P(n) = a_1 a_2 \cdots a_k$. If n and $n + 23$ differ in any digit preceding the hundreds place (where leading digits of 0 are allowed if necessary to compare these digits), then one of the digits of $n+23$ must be 0, which means $P(n+23) = 0$, and thus n would not be a solution to the given equation.

Otherwise, suppose $k > 3$, and let $n + 23 = \underline{a_1}\,\underline{a_2}\,\ldots\,\underline{a_{k-3}}\,\underline{b_{k-2}}\,\underline{b_{k-1}}\,\underline{b_k}$, where b_{k-2}, b_{k-1}, b_k may differ from a_{k-2}, a_{k-1}, a_k, respectively. The given equation implies

$$23 = P(n + 23) - P(n) = (b_{k-2}b_{k-1}b_k - a_{k-2}a_{k-1}a_k) \prod_{j=1}^{k-3} a_j.$$

Because the product is a divisor of 23 and 23 is prime, it follows that the product is equal to 1 and therefore, $a_j = 1$ for all $1 \leq j < k - 2$. This means that the number $n = \underline{a_{k-2}}\,\underline{a_{k-1}}\,\underline{a_k}$ with $n + 23 = \underline{b_{k-2}}\,\underline{b_{k-1}}\,\underline{b_k}$ would be a strictly smaller solution to the given equation. Therefore it suffices to find all solutions for n with at most three digits.

All solutions to $P(n + 23) = P(n) + 23$ will be of the form $n = 34$, $n = 51$, $n = 11\ldots134$, or $n = 11\ldots151$. The least such solution is $n = 34$.

Problem 3. Compute the least integer value of the function

$$f(x) = \frac{x^4 - 6x^3 + 2x^2 - 6x + 2}{x^2 + 1}$$

whose domain is the set of all real numbers.

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Solution 3. Use polynomial long division to rewrite $f(x)$ as

$$f(x) = x^2 - 6x + 1 + \frac{1}{x^2+1}.$$

The quadratic function $x^2 - 6x + 1 = (x-3)^2 - 8$ has a minimum of -8, achieved at $x = 3$. The "remainder term" $\frac{1}{x^2+1}$ is always positive. Thus $f(x) > -8$ for all x, so any integer value of $f(x)$ must be at least -7.

When $x = 3$, the remainder term is less than 1, so $f(3)$ is less than -7. But $f(4) = -\frac{34}{5} > -7$, so there must be some value of x between 3 and 4 for which $f(x) = -7$, so the least integer value of $f(x)$ is $\mathbf{-7}$. The reader may note that $f(x) = -7$ when $x \approx 2.097$ and $x \approx 3.970$.

Problem 4. Suppose that noncongruent triangles ABC and XYZ are given such that $AB = XY = 10$, $BC = YZ = 9$, and $\text{m}\angle CAB = \text{m}\angle ZXY = 30^\circ$. Compute $[ABC] + [XYZ]$.

Solution 4. Because triangles ABC and XYZ are noncongruent yet have two adjacent sides and an angle in common, the two triangles are the two possibilities in the ambiguous case of the Law of Sines. Without loss of generality, let triangle ABC have obtuse angle C and triangle XYZ have acute angle Z so that $\text{m}\angle C + \text{m}\angle Z = 180^\circ$. Place triangle ABC so that B and Y coincide, and C and Z coincide. Because $\text{m}\angle C$ and $\text{m}\angle Z$ add up to 180°, it follows that points X, Z, and A all lie on the same line. The two triangles together then form $\triangle ABX$, where $\text{m}\angle BAX = \text{m}\angle BXA = 30^\circ$ and $BX = AB = 10$. Therefore the sum of the areas of the two triangles is equal to the area of triangle ABX, which is $\frac{1}{2} \cdot 10 \cdot 10 \cdot \sin(120^\circ) = \frac{5 \cdot 10 \cdot \sqrt{3}}{2} = \mathbf{25\sqrt{3}}$.

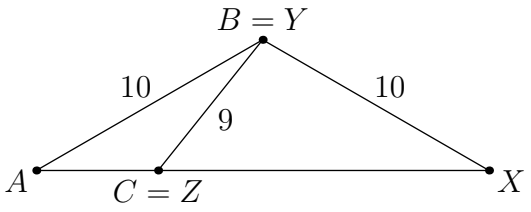


Figure not drawn to scale.

Alternate Solution: As explained above, let $\triangle ABC$ have obtuse angle C and $\triangle XYZ$ have acute angle Z. By the Law of Sines, $\sin(\angle C) = \sin(\angle Z) = \frac{5}{9}$. This implies $\text{m}\angle XYZ = \frac{5\pi}{6} - \arcsin\left(\frac{5}{9}\right)$ and $\text{m}\angle ABC = \arcsin(\frac{5}{9}) - \frac{\pi}{6}$. The areas of the triangles are $[XYZ] = \frac{1}{2} \cdot 10 \cdot 9 \cdot \sin\left(\frac{5\pi}{6} - \arcsin(\frac{5}{9})\right)$ and $[ABC] = \frac{1}{2} \cdot 10 \cdot 9 \cdot \sin\left(\arcsin(\frac{5}{9}) - \frac{\pi}{6}\right)$. By the angle subtraction rule, it follows that

$$\sin\left(\frac{5\pi}{6} - \arcsin\left(\frac{5}{9}\right)\right) = \sin\left(\frac{5\pi}{6}\right)\cos\left(\arcsin\left(\frac{5}{9}\right)\right) - \cos\left(\frac{5\pi}{6}\right)\sin\left(\arcsin\left(\frac{5}{9}\right)\right) \text{ and}$$
$$\sin\left(\arcsin\left(\frac{5}{9}\right) - \frac{\pi}{6}\right) = \sin\left(\arcsin\left(\frac{5}{9}\right)\right)\cos\left(\frac{\pi}{6}\right) - \cos\left(\arcsin\left(\frac{5}{9}\right)\right)\sin\left(\frac{\pi}{6}\right).$$

The sum of the two sines is $\sin\left(\arcsin(\frac{5}{9})\right)\left(\cos(\frac{\pi}{6}) - \cos(\frac{5\pi}{6})\right) = \frac{5}{9} \cdot \sqrt{3}$ because $\sin(\frac{\pi}{6}) = \sin(\frac{5\pi}{6})$. Finally, the sum of the areas of the two triangles is $\frac{1}{2} \cdot 10 \cdot 9 \cdot \frac{5}{9} \cdot \sqrt{3} = 25\sqrt{3}$.

Problem 5. The mean, median, and unique mode of a list of positive integers are three consecutive integers in some order. Compute the least possible sum of the integers in the original list.

Solution 5. One possible list is 1, 1, 3, 7, which has mode 1, median 2, and mean 3. The sum is $1+1+3+7 = 12$. A list with fewer than four numbers cannot produce a median and unique mode that are distinct from each other. To see this, first note that a list with one number has the same median and mode. In a list with two numbers, the mode is not unique if the numbers are different, and if the numbers are the same, the median and mode are equal. In a list of three numbers with a unique mode, the mode must occur twice. Hence the

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mode is equal to the middle number of the three, which is the median. Thus a list with a median and unique mode that are different from each other must contain at least four numbers.

Now suppose that a list satisfying the given conditions sums to less than 12. The mean must be greater than 1, and because the list contains at least four numbers, the mean must be exactly 2. The median must also be greater than 1, and if the mode is 4, then the sum must be greater than 12. Thus it remains to determine if a mean of 2 with mode 1 and median 3 can be achieved with a list of four or five positive integers. However, having two 1s in the list and a median of 3 forces the remaining numbers in each case to have a sum too large for a mean of 2. The least possible sum is therefore **12**.

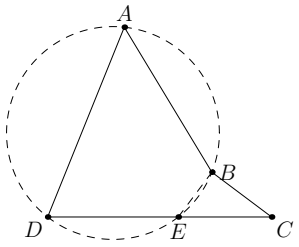
Problem 6. David builds a circular table; he then carves one or more positive integers into the table at points equally spaced around its circumference. He considers two tables to be the same if one can be rotated so that it has the same numbers in the same positions as the other. For example, a table with the numbers 8, 4, 5 (in clockwise order) is considered the same as a table with the numbers 4, 5, 8 (in clockwise order), but both tables are different from a table with the numbers 8, 5, 4 (in clockwise order). Given that the numbers he carves sum to 17, compute the number of different tables he can make.

Solution 6. The problem calls for the number of ordered partitions of 17, where two partitions are considered the same if they are cyclic permutations of each other. Because 17 is prime, each ordered partition of 17 into n parts will be a cyclic permutation of exactly n such partitions (including itself), unless $n = 17$. (If $n = 17$, then all the numbers are 1s, and there is exactly one table David can make.) By the sticks and stones method, the number of ordered partitions of 17 into n nonzero parts is $\binom{16}{n-1}$, and this overcounts the number of tables by a factor of n, except when $n = 17$. Thus the number of possible tables is

$$1 + \sum_{n=1}^{16} \binom{16}{n-1} \cdot \frac{1}{n} = 1 + \sum_{n=1}^{16} \binom{17}{n} \cdot \frac{1}{17} = 1 + \frac{2^{17} - 2}{17} = \mathbf{7711}.$$

Problem 7. In quadrilateral $ABCD$, $\mathrm{m}\angle B + \mathrm{m}\angle D = 270^\circ$. The circumcircle of $\triangle ABD$ intersects $\overline{CD}$ at point E, distinct from D. Given that $BC = 4$, $CE = 5$, and $DE = 7$, compute the diameter of the circumcircle of $\triangle ABD$.

Solution 7. Note that $\mathrm{m}\angle A + \mathrm{m}\angle C = 90^\circ$ in quadrilateral $ABCD$. Because quadrilateral $ABED$ is cyclic, it follows that $\mathrm{m}\angle ADE + \mathrm{m}\angle ABE = 180^\circ$. Moreover, because $\mathrm{m}\angle ABE + \mathrm{m}\angle EBC + \mathrm{m}\angle ADE = 270^\circ$, it follows that $\angle EBC$ is a right angle. Thus $BE = \sqrt{CE^2 - BC^2} = \sqrt{5^2 - 4^2} = 3$. Let $\mathrm{m}\angle BEC = \theta$; then $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$.



Applying the Law of Cosines to $\triangle BED$ yields

$$BD^2 = 3^2 + 7^2 - 2 \cdot 3 \cdot 7 \cos(180^\circ - \theta) = 3^2 + 7^2 + 2 \cdot 3 \cdot 7 \cos\theta = \frac{416}{5}.$$

Thus $BD = \frac{4\sqrt{26}}{\sqrt{5}}$. Let R be the circumradius of $\triangle ABD$ and $\triangle BED$. Then the requested diameter is $2R$, and

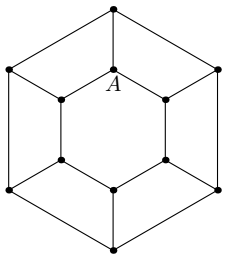
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applying the Law of Sines to $\triangle BED$ yields

$$2R = \frac{BD}{\sin(180^\circ - \theta)} = \frac{BD}{\sin\theta} = \frac{4\sqrt{26}}{\sqrt{5}} \cdot \frac{5}{4} = \boxed{\sqrt{130}}.$$

Alternate Solution: Proceed as above to conclude that $\angle EBC$ is a right angle and $BE = 3$. Extend $\overline{BC}$ past B to intersect the circle a second time at F. By Power of a Point from C, it follows that $CE \cdot CD = CB \cdot CF$, from which $CF = \frac{5 \cdot 12}{4} = 15$, and $BF = 11$. Because $\angle EBF$ is also a right angle, it follows that $\overline{EF}$ is a diameter of the circle and that $EF = \sqrt{3^2 + 11^2} = \sqrt{130}$.

Problem 8. Suppose that Xena traces a path along the segments in the figure shown, starting and ending at point A. The path passes through each of the eleven vertices besides A exactly once, and only visits A at the beginning and end of the path. Compute the number of possible paths Xena could trace.



Solution 8. Count the number of complete paths that pass through all vertices exactly once (such a path is called a *Hamiltonian path*). The set of vertices can be split into two rings:

$$\mathcal{I} = \{A_1, A_2, \ldots, A_6\} \text{ (i.e., the inner ring)}, \quad \mathcal{O} = \{B_1, B_2, \ldots, B_6\} \text{ (i.e., the outer ring)}.$$

where $A_1 = A$. The two rings are connected by the edges $E = \{A_1B_1, A_2B_2, \ldots, A_6B_6\}$. Each vertex in the figure has exactly three edges joining it with the neighboring vertices. Also note that any closed loop must use exactly two edges (out of three) for each vertex.

Further note that a loop must use at least one edge from E to move from one ring to the other. Consider two cases: the loop uses all six edges from E, or it uses some but not all of them.

If all edges from E are used, there are two possible undirected loops. It is not possible to use both edges A_1A_2 and B_1B_2, so either A_1A_2 or B_1B_2 will be used. This choice determines how the entire loop is constructed.

If not all edges from E are used, then there must be some i for which the loop uses the edge A_iB_i and does not use $A_{i+1}B_{i+1}$ (where A_jB_j represents $A_{j-6}B_{j-6}$ if $7 \le j \le 12$). Because A_{i+1} is only connected to three other vertices, the loop must use A_iA_{i+1} and $A_{i+1}A_{i+2}$, and similarly must use B_iB_{i+1} and $B_{i+1}B_{i+2}$. This also precludes using $A_{i+2}B_{i+2}$, because doing so would close the loop before it visits all 12 vertices. Therefore the loop must also use $A_{i+2}A_{i+3}$ and $B_{i+2}B_{i+3}$, which now precludes using $A_{i+3}B_{i+3}$. This continues to force the structure of the loop until it closes by using $A_{i+5}B_{i+5} = A_{i-1}B_{i-1}$. Hence the loop must use exactly two edges from E, and they must be consecutive: $A_{i-1}B_{i-1}$ and A_iB_i. There are 6 ways to choose those two consecutive edges, so there are 6 possible undirected loops in this case.

The forced path is a loop, and the only way the given conditions are satisfied if $A_{i+k} = A_{i-1}$ and $B_{i+k} = B_{i-1}$. Hence the loop must use (precisely) two consecutive edges from E: $A_{i-1}B_{i-1}$ and A_iB_i. There are 6 ways to choose two consecutive edges, so there are 6 possible undirected loops in this case.

Each undirected loop can be traced in two ways, and thus the number of ways for Xena to trace the path is $(6 + 2) \cdot 2 = \mathbf{16}$.

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Problem 9. Let $i = \sqrt{-1}$. The complex number $z = -142 + 333\sqrt{5}i$ can be expressed as a product of two complex numbers in multiple different ways, two of which are $\left(57 - 8\sqrt{5}i\right)\left(-6 + 5\sqrt{5}i\right)$ and $\left(24 + \sqrt{5}i\right)\left(-3 + 14\sqrt{5}i\right)$. Given that $z = -142 + 333\sqrt{5}i$ can be written as $\left(a + b\sqrt{5}i\right)\left(c + d\sqrt{5}i\right)$, where a, b, c, and d are *positive* integers, compute the lesser of $a + b$ and $c + d$.

Solution 9. Multiply each of the given parenthesized expressions by its complex conjugate to obtain

$$\begin{aligned} 142^2 + 5 \cdot 333^2 &= \left(57^2 + 5 \cdot 8^2\right)\left(6^2 + 5 \cdot 5^2\right) \\ &= \left(24^2 + 5 \cdot 1^2\right)\left(3^2 + 5 \cdot 14^2\right) \\ &= \left(a^2 + 5b^2\right)\left(c^2 + 5d^2\right). \end{aligned}$$

The expression on the second line is equal to $581 \cdot 989 = 7 \cdot 83 \cdot 23 \cdot 43$ (one can perhaps factor 989 a little faster by noting that 23 divides $6^2 + 5 \cdot 5^2 = 7 \cdot 23$ but not 581, so it must divide 989). Thus $a^2 + 5b^2$ and $c^2 + 5d^2$ must be a factor pair of this number. It is not possible to express 1, 7, 23, 43, or 83 in the form $x^2 + 5y^2$ for integers x, y.

Let $N = a^2 + 5b^2$, and without loss of generality, assume that 7 divides N. From the above analysis, N must be $7 \cdot 23$, $7 \cdot 43$, or $7 \cdot 83$. By direct computation of checking all positive integers b less than $\sqrt{\frac{N}{5}}$, the only possibilities for (a, b) are:

- when $N = 7 \cdot 23$, either $(9, 4)$ or $(6, 5)$;
- when $N = 7 \cdot 43$, either $(16, 3)$ or $(11, 6)$; and
- when $N = 7 \cdot 83$, either $(24, 1)$ or $(9, 10)$.

Next, observe that

$$\frac{-142 + 333\sqrt{5}i}{a + b\sqrt{5}i} = \frac{(-142a + 1665b) + (333a + 142b)\sqrt{5}i}{N}$$

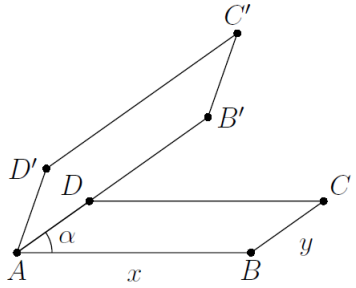
must equal $c + d\sqrt{5}i$, so N must divide $-142a + 1665b$ and $333a + 142b$. But

- 7 does not divide $333 \cdot 9 + 142 \cdot 4$ or $333 \cdot 6 + 142 \cdot 5$;
- 43 does not divide $333 \cdot 16 + 142 \cdot 3$; and
- 83 does not divide $333 \cdot 9 + 142 \cdot 10$.

Thus the only candidates are $(a, b) = (11, 6)$ and $(a, b) = (24, 1)$. Note that $(24, 1)$ yields the second factorization given in the problem statement, which has a negative real part in one of its factors. Thus the only remaining candidate for (a, b) is $(11, 6)$, which yields $(c, d) = (28, 15)$, thus the answer is $11 + 6 = \mathbf{17}$.

Problem 10. Parallelogram $ABCD$ is rotated about A in the plane, resulting in $AB'C'D'$, with D on $\overline{AB'}$. Suppose that $[B'CD] = [ABD'] = [BCC']$. Compute $\tan \angle ABD$.

Solution 10. Let $AB = x$, $BC = y$, and $\mathrm{m}\angle A = \alpha$.



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It then follows that $[ABD'] = \frac{xy \sin 2\alpha}{2}$ and $[B'CD] = \frac{x(x-y) \sin \alpha}{2}$.

Because $\overline{BC}$, $\overline{AB'}$, and $\overline{D'C'}$ are all parallel, it follows that $\triangle BCC'$ and $\triangle BCD'$ have the same height with respect to base $\overline{BC}$, and thus $[BCC'] = [BCD']$. Therefore $[BCD'] = [ABD']$, and it follows that triangles BCD' and ABD' have the same height with respect to base $\overline{BD'}$. Thus A and C are equidistant from $\overleftrightarrow{BD'}$.

This implies that the midpoint of $\overline{AC}$ lies on $\overleftrightarrow{BD'}$. But $\overleftrightarrow{BD}$ also passes through the midpoint of $\overline{AC}$ by parallelogram properties, so it follows that D must lie on $\overline{BD'}$. This implies that $[ABD']$ must also equal $\frac{y^2 \sin \alpha}{2} + \frac{xy \sin \alpha}{2} = \frac{(xy+y^2) \sin \alpha}{2}$.

This implies that $x(x-y) \sin \alpha = xy \sin 2\alpha = (xy + y^2) \sin \alpha$, which implies $x : y = \sqrt{2} + 1$ and $\sin \alpha = \cos \alpha = \frac{\sqrt{2}}{2}$. Finally, from right triangle $D'AB$ with legs in the ratio $1 : \sqrt{2}+1$, it follows that $\tan(\angle ABD) = \tan(\angle ABD') = \mathbf{\sqrt{2} - 1}$.

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